

Functional Analysis — Homework 6

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Problem 1. Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be an analytic function, and let

$$f(x, y) = u(x, y) + iv(x, y)$$

be its decomposition into real and imaginary parts. Consider the limit definition of the complex derivative at a point $z = x + iy$:

$$f'(z) = \lim_{w \rightarrow 0} \frac{f(z+w) - f(z)}{w}.$$

- (a) Evaluate this limit by letting $w \rightarrow 0$ strictly along the real axis. Express the result in terms of the partial derivatives of u and v .
- (b) Now evaluate the same limit by letting $w \rightarrow 0$ strictly along the imaginary axis.

Since f is analytic, the limit must be independent of the path. Equate your results from (a) and (b) to prove the Cauchy-Riemann equations:

$$u_x = v_y, \quad u_y = -v_x.$$

Problem 2. Use the limit definition to show that $f(z) = \bar{z}$ is not complex differentiable anywhere, and $g(z) = |z|^2$ is not complex differentiable at any $z \neq 0$, but is complex differentiable at $z = 0$.

Problem 3. Fix a small, positive radius $r < 1$ and let γ_1 and γ_2 be the positively oriented circles of radius r centered at $z = 1$ and $z = 2$, respectively:

$$\gamma_1 = 1 + re^{it}, \quad \gamma_2(t) = 2 + re^{it}, \quad t \in [0, 2\pi].$$

Compute the following line integrals for $j = 1, 2$:

$$(a) \int_{\gamma_j} \frac{dz}{1-z} \quad (b) \int_{\gamma_j} \frac{dz}{2-z} \quad \int_{\gamma_j} \frac{dz}{(1-z)^2} \quad (d) \int_{\gamma_j} \frac{dz}{(2-z)(1-z)}.$$

Hint: for item (d), partial fractions can immediately reduce it to (a) and (b).

Problem 4. Consider the linear operator $A : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ represented in the standard basis by the matrix

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Note that the spectrum of A is $\sigma(A) = \{1, 2\}$.

- (a) Find the kernel of $(A-1)^2$ and the kernel of $A-2$. These are the generalized eigenspaces of A for the eigenvalues 1 and 2.
- (b) Compute the resolvent matrix $R(z) = (A - z)^{-1}$ for $z \notin \sigma(A)$. Simplify the rational functions in each entry completely.
- (c) For each eigenvalue $\lambda_j = j, j = 1, 2$ compute the operator-valued integral

$$P_j = -\frac{1}{2\pi i} \int_{\gamma_j} R(z) dz,$$

where γ_j is a small positively oriented circle enclosing only λ_j .

- (d) Show that P_1 and P_2 are projections, that their images are the generalized eigenspaces of A ,

$$\text{Im}(P_1) = \ker(A - 1)^2, \quad \text{Im}(P_2) = \ker(A - 2),$$

and that they complement each other, meaning:

$$P_1 + P_2 = 1, \quad P_1 P_2 = 0.$$

P_1 and P_2 are called *Riesz projections* associated with A .